ROHIT KALE

AI Engineer & Architect | Computer Vision & GenAI | AWS Certified Solutions Architect

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PROFILE

AI Engineer & Architect with 4.5+ years of experience building production-grade computer vision, NLP and full-stack AI systems. At Cellstrat AI, built the **UBall** basketball analytics platform: multi-camera player tracking, shot detection and dual-angle fusion using YOLO, SAM and ByteTrack, Gemini VLM game analysis, and deployment from NVIDIA Jetson edge devices to AWS and GCP. Built **NxChat**, an enterprise RAG chat system on LangChain and AWS OpenSearch with multi-model support (AWS Bedrock + OpenAI). Specialized in end-to-end ML pipelines from model training to production serving using Python, FastAPI, PyTorch, Docker and cloud infrastructure (AWS & GCP). AWS Certified Solutions Architect with an MS in Artificial Intelligence.

TECHNICAL SKILLS

Architecture: event-driven & batch ML pipelines, microservices, edge-to-cloud ingestion, LLM gateways with provider fallback, RAG reference designs, API design, multi-tenant RBAC, cost & security guardrails

Computer Vision: YOLOv8 / YOLO11 / YOLO26, RF-DETR, SAM 2 / SAM 3, ByteTrack, DeepSORT, RTMPose, PARSeq OCR, Kalman / RTS smoothing, homography & multi-camera calibration, OpenCV, Detectron2

GenAI & Agents: OpenAI, Gemini (VLM), Claude, AWS Bedrock; LangChain, LangGraph, Google ADK, MCP; RAG (OpenSearch k-NN, Qdrant, sqlite-vec); OpenAI Realtime API, LiveKit voice agents; Vertex AI fine-tuning

ML & Evaluation: PyTorch, Ultralytics, scikit-learn, interpretable models, blind / held-out evaluation, dataset building & annotation tooling, reinforcement learning (DQN)

Backend: Python, FastAPI, asyncio, Celery + Redis, SQLAlchemy (async), Alembic, Pydantic, WebSocket, REST, Pytest

Cloud & MLOps: AWS (EC2 GPU / spot, S3, Lambda, Step Functions, ECS / ECR, OpenSearch, CloudWatch), GCP (Vertex AI, Cloud Run Jobs, Cloud Functions, Workflows, GCS), Docker, GitHub Actions, NVIDIA Jetson, Cloudflare R2 / Stream

Data & Frontend: PostgreSQL, Supabase (Auth, RLS), Redis, Firebase; React, Next.js, TypeScript, Vite, Tailwind, shadcn/ui, Flutter

PROFESSIONAL EXPERIENCE

AI/ML Engineer Cellstrat AI

Apr 2025 – Present

UBALL: AI BASKETBALL ANALYTICS (LATEST)

- Designed and built a **4-camera game-understanding pipeline**: YOLO26 / RF-DETR detection, RTMPose, jersey OCR, ByteTrack, cross-camera identity fusion and a ball-possession state machine that emits shot, rebound and turnover events. Architecture runs from Jetson edge capture through S3 and AWS GPU batch jobs.
- Evaluated the system against human-annotated ground truth on held-out games, and retrained the far-angle model on production frames to improve shot detection.
- Built Gemini VLM play narration, a Vertex AI fine-tuning pipeline on GCP, and the FastAPI / Supabase / Celery product backend.

CELLASSIST: VOICE AI FOR HEALTHCARE

- Built a voice AI agent for patient triage that connects Exotel telephony WebSockets to the **OpenAI Realtime API** for speech-to-speech calls in multiple languages.
- Kept conversations natural with audio buffering, noise suppression and interruption handling. Deployed on AWS Mumbai for low caller latency.

NXCHAT: ENTERPRISE RAG CHAT

- Built an enterprise chat system with **RAG** on LangChain and AWS OpenSearch vector search, routing across AWS Bedrock and OpenAI models.
- Added dataset management, file filtering and a versioned FastAPI REST API, deployed on AWS EC2 / S3 with a Pytest suite.

REXAI: GENAI FILE CLASSIFICATION

- Built a scalable classification backend that combines lexical techniques with **LLM-based classification** to keep high-volume work accurate and cheap.

- Ran processing asynchronously on AWS Lambda + Step Functions with OpenSearch indexing for high-volume workloads.

CELLBOT: AI CHATBOT & REPORTS

- Built the FastAPI backend (Poetry, Pydantic, WeasyPrint) that generates PDF reports from conversations.
- Built the Next.js frontend with Clerk authentication, shadcn/ui, real-time markdown rendering and conversation history.

Software Engineer **Independent Consultant** Aug 2023 – Apr 2025

- Built domain-specific ChatGPT chatbots and PayPal subscription billing (Supabase) for clients' FlutterFlow apps.
- Automated an end-to-end reading service with n8n + OpenAI: Razorpay payment, form extraction, generated documents and Gmail delivery, with retries and error handling.
- Containerised AI services with Docker and deployed them on AWS (Lambda, S3, CloudWatch) with GitHub Actions CI/CD.

Analyst **TheMathCompany, Bangalore** Feb 2022 – Aug 2023

- Built data pipelines for ML model training and evaluation for retail and BFSI clients, plus API-driven automation for client data workflows.
- Built dashboards that tracked system performance and data-quality metrics for stakeholders.

EDUCATION & CERTIFICATION

AWS Certified Solutions Architect – Associate, Amazon Web Services 2025

MS, Artificial Intelligence, Liverpool John Moores University 2021 – 2023

Post Graduate Diploma, Machine Learning & AI, IIIT Bangalore 2021 – 2022

B.Tech, Visvesvaraya National Institute of Technology (VNIT), Nagpur 2016 – 2020